Project: Worked examples
Title: ISO 286 limits and fits
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

Hole and shaft limits for a nominal size and tolerance classes, the resulting fit (clearance, transition or interference) and its extreme values, from the ISO 286-1 formulae.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Standard tolerance factor at the geometric mean D of the size step | i = 0.45·D^(1/3) + 0.001·D (µm) |
| Width of the tolerance band for the grade | ITn = multiplier × i |
| Fundamental deviations by letter from the published formulae, and hole deviations by the general rule | EI = −es (A–H); ES = −ei + Δ (K–ZC) |
| Limits of size of the hole and the shaft | limit = nominal + deviation |
| Fit extremes, the largest and smallest clearance (negative is interference) | $c_{max}$ = ES − ei; $c_{min}$ = EI − es |
| Rolling-bearing bore mode | a bearing bore is not an ISO 286 hole; it has its own tolerance (ISO 492) lying below nominal, so ES and EI are entered from the bearing record instead of being computed |

Basis: Bases of tolerances, deviations and fits — tolerance grades and fundamental deviations[1](§4, §5); Tables of standard tolerance classes and limit deviations (not reproduced)[2].

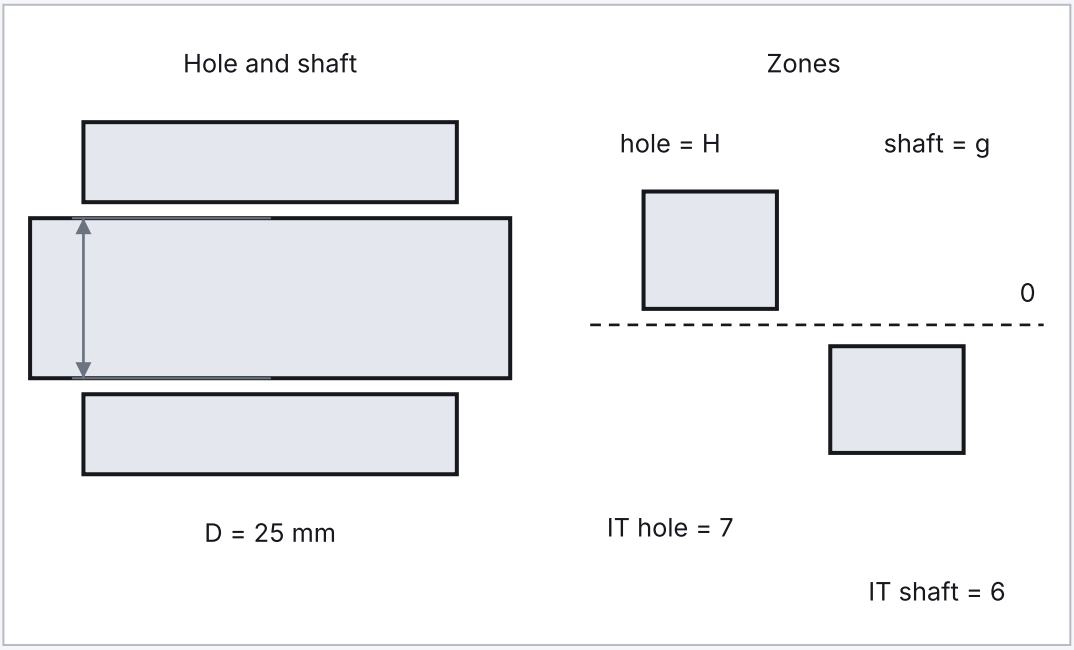



Figure 1 — A hole and its shaft of one nominal size D, and their tolerance zones about the zero line (hole above, shaft below as drawn; the letters place them). Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| D | Nominal size | 25 | mm | drawing: DRG-SHAFT-12 rev A | 1 … 500 mm |
| $hole_{kind}$ | Hole: an ISO 286 class, or a rolling-bearing bore (ES and EI from the bearing record, ISO 492) | ISO 286 hole class | | default | any |
| $hole_{letter}$ | Hole fundamental deviation letter | H | | spec: DRG-SHAFT-12 note 3: H7/g6 | any |
| $hole_{grade}$ | Hole IT grade | 7 | | spec: DRG-SHAFT-12 note 3 | 5 … 11 |
| $shaft_{letter}$ | Shaft fundamental deviation letter | g | | spec: DRG-SHAFT-12 note 3 | any |
| $shaft_{grade}$ | Shaft IT grade | 6 | | spec: DRG-SHAFT-12 note 3 | 4 … 11 |

## 3. Assumptions

A1. IT values and fundamental deviations are the unrounded results of the ISO 286-1 formulae at the geometric mean of the size step. (The tabulated values apply the standard's own rounding and may differ by about 1 µm for the clearance and transition letters (a to n). For the interference letters p to zc the tables also carry the standard's own increments and sub-steps, so the formula value can be several µm below the tabulated one (p6 at 450 mm: 62.2 against 68 µm), and the sheet warns; enter overrides from a released reference record where the tabulated value governs.)
A2. Hole deviations follow the general rule from the shaft deviations, with Δ = ITn − IT(n−1) where the standard applies it. (Table-specific departures in the standard are not reproduced; use the overrides where they govern.)

## 4. Working

[7] $D_{step}$ = size step (lower, upper] [1](§4)
= (18, 30] mm
= 23.238 mm
Geometric mean of the size step

[8] i = 0.45·D^(1/3) + 0.001·D [1](§5)
= D = 23.238 mm
= 1.3074 µm
Standard tolerance unit

[9] $IT_{hole}$ = IT7 [1](§5)
= 16 × i
= 20.918 µm
Hole tolerance

[10] $IT_{shaft}$ = IT6 [1](§5)
= 10 × i
= 13.074 µm
Shaft tolerance

[11] es = fundamental deviation 'g' [1](§5)
= -2.5·D^0.34
= -7.2853 µm
Shaft upper deviation

[12] $ei = es - IT_{shaft}$ [1](§5)
$= -7.2853 - 13.074$
$= -20.359\ \mu m$
Shaft lower deviation
[13] $ES = EI + IT_{hole}$ [1](§5)
$= 0 + 20.918$
$= 20.918\ \mu m$
Hole upper deviation
[14] $EI$ = fundamental deviation 'H' [1](§5)
= -es of the same letter
$= 0\ \mu m$
Hole lower deviation
[15] $D_{hole,max} = D + ES$
$= 25\ mm + 20.918\ \mu m$
$= 25.021\ mm$
Hole maximum
[16] $D_{hole,min} = D + EI$
$= 25\ mm + 0\ \mu m$
$= 25\ mm$
Hole minimum
[17] $d_{shaft,max} = D + es$
$= 25\ mm + -7.2853\ \mu m$
$= 24.993\ mm$
Shaft maximum
[18] $d_{shaft,min} = D + ei$
$= 25\ mm + -20.359\ \mu m$
$= 24.98\ mm$
Shaft minimum
[19] $c_{max} = ES - ei$
$= 20.918\ \mu m - (-20.359\ \mu m)$
$= 41.277\ \mu m$
Maximum clearance (negative: minimum interference)
[20] $c_{min} = EI - es$
$= 0\ \mu m - (-7.2853\ \mu m)$
$= 7.2853\ \mu m$
Minimum clearance (negative: maximum interference)
[21] $i_{max} = es - EI$
$= (-7.2853\ \mu m) - 0\ \mu m$
$= -7.2853\ \mu m$
Maximum interference
[22] $i_{min} = ei - ES$
$= (-20.359\ \mu m) - 20.918\ \mu m$
$= -41.277\ \mu m$
Minimum interference (negative: maximum clearance)
[23] fit = fit class
= H7/g6
= "clearance"
Fit kind: clearance, transition or interference
[24] $c_{mean} = (c_{max} + c_{min}) / 2$
$= (41.277\ \mu m + 7.2853\ \mu m) / 2$
$= 24.281\ \mu m$
Mean clearance

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $IT_{hole}$ | 20.918 µm | Hole tolerance |
| $IT_{shaft}$ | 13.074 µm | Shaft tolerance |
| ES | 20.918 µm | Hole upper deviation |
| EI | 0 µm | Hole lower deviation |
| es | -7.2853 µm | Shaft upper deviation |
| ei | -20.359 µm | Shaft lower deviation |
| $D_{hole,max}$ | 25.021 mm | Hole maximum |
| $D_{hole,min}$ | 25 mm | Hole minimum |
| $d_{shaft,max}$ | 24.993 mm | Shaft maximum |
| $d_{shaft,min}$ | 24.98 mm | Shaft minimum |
| fit | "clearance" | Fit kind |
| $c_{max}$ | 41.277 µm | Maximum clearance |
| $c_{min}$ | 7.2853 µm | Minimum clearance |
| $i_{max}$ | -7.2853 µm | Maximum interference |
| $i_{min}$ | -41.277 µm | Minimum interference |
| $c_{mean}$ | 24.281 µm | Mean clearance |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Fit classified | H7/g6 | clearance | 0 | |

## 7. Applicability

WARNING — input.hole_kind: $hole_{kind}$ (Hole: an ISO 286 class, or a rolling-bearing bore (ES and EI from the bearing record, ISO 492)) was not entered; the template default 'ISO 286 hole class' is used.

## 8. Verification

```
Template mech.iso286-fit@1.4.0 Content hash 879a741a1c209919 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

## 9. References

1. International Organization for Standardization. Geometrical Product Specifications (GPS)—ISO Code System for Tolerances on Linear Sizes—Part 1: Basis of Tolerances, Deviations and Fits. ISO 286-1:2010. ISO; 2010.
2. International Organization for Standardization. Geometrical Product Specifications (GPS)—ISO Code System for Tolerances on Linear Sizes—Part 2: Tables of Standard Tolerance Classes and Limit Deviations for Holes and Shafts. ISO 286-2:2010. ISO; 2010.